Disaster Recovery and Restoration

Project Report

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This project report documents the corporate office infrastructure disaster recovery and systems restoration project. The target audience is: corporate executive management.

1. Executive Summary

This project was initiated in response to a major environmental disaster in 2026, when severe regional rainstorms caused widespread municipal flooding across Canada, impacting every property in the local district and severely inundating the commercial offices of DTEK Consulting Services Ltd. The flood caused extensive physical damage to the facility and destroyed core server hardware, which compromised primary infrastructure services and eliminated high-availability redundancy across the corporate network.

To restore business continuity and physical operations, a multi-phased disaster recovery program was managed and funded by DTEK Consulting Services Ltd. Physical facility restoration required strict project sequencing and oversight of multiple specialized trade contractors. Initial emergency efforts focused on environmental containment, structural drying, and the removal of flood-damaged drywall, insulation, and collapsed ceilings in affected areas. Following comprehensive general building and electrical safety inspections, the space underwent specialized antimicrobial treatments for mold prevention before structural repairs, wall reconstructions, and commercial carpet replacement were completed.

Concurrently, a comprehensive systems restoration phase was executed to rebuild lost IT systems server capacity and re-establish network stability. Replacement enterprise server hardware was procured, configured, and integrated to re-host all affected business workloads. Technical remediation focused on restoring core directory services, eliminating single points of failure, and validating failover redundancy through a structured test-driven framework.

The completed project restored the company’s office facility to operational standards, delivering a modern, highly available technical infrastructure engineered for disaster resilience.

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2. Project Overview

The disaster recovery project began immediately following a severe weather event in 2026, when torrential rainstorms caused widespread district flooding across Canada. Commercial and residential properties across the local area suffered severe water inundation, including the corporate offices of DTEK Consulting Services Ltd. Immediate emergency response operations were launched to halt active water entry, mitigate structural damage, and protect server infrastructure supporting business-critical systems, such as accounting, bookkeeping, corporate tax records, payroll, and marketing.

Initial physical containment was caried out continuously during the first two weeks of the disaster. Emergency water extraction was performed manually across affected areas using standard handheld mops and wet vacuums to limit deep saturation of subflooring and carpets. To address the primary source of incoming water, emergency roof access was established to clear severely blocked drainage systems and gutters by hand, reducing structural water infiltration and saving the facility from catastrophic total loss. Simultaneously, heavy roof leakage in a nearby building section led to a ceiling collapse, requiring immediate debris removal, structural gutting, and extraction of saturated overhead insulation to prevent further ceiling failure and structural decay.

Following initial containment, building assessments were conducted by structural, electrical, and plumbing inspectors. These inspections established an action plan to guide structural and technical recovery. Based on inspector findings, trade-scale repairs were scheduled and initiated:

- Environmental Containment & Water Extraction: Rapid removal of standing water and continuous extraction across affected flooring zones.
- Selective Structural Demolition: Cutting and removing flood-damaged drywall assemblies and saturated wall-cavity insulation to prepare for structural drying.
- Antimicrobial Mold Remediation: Comprehensive chemical treatments applied to exposed wall studs, subfloors, and high-risk moisture entry points to prevent mold growth.
- Specialized Trade Inspections: Detailed electrical and plumbing assessments to check structural integrity and ensure safe re-energization.

- Contractor Sequencing & Facility Reconstruction: Coordinating scheduling of wall reconstruction, ceiling restoration, and carpet installation based on trade availability during the crisis recovery period.
- IT Systems Restoration & Hardware Replacement: Procuring replacement server hardware to rebuild flood-destroyed host infrastructure; recover critical business data (financial, tax, payroll, marketing); and engineer purposeful system redundancy to ensure fault tolerance and high-availability failover.

Through coordinated building repairs and IT system restoration, DTEK Consulting Services Ltd. successfully stabilized the facility, eliminated environmental hazards, and brought core IT services back online.

3. Project Outcomes

3.1 Operational and Facility Restoration

The disaster recovery project successfully restored core business operations and physical premises to full functional capacity, establishing a clean, reliable baseline across both the facility and IT environment.

- Emergency site stabilization and flood mitigation: Immediate, hands-on response halted active water ingress, prevented further structural deterioration, and protected core server infrastructure during the initial flooding event.
- Physical office space fully remediated: Intensive physical cleanup, structural gutting, drywall replacement, fresh painting, and carpet installations were completed across all flood-affected areas, returning the workplace to safe, functional, and professionally restored condition.
- Building safety and environmental certification: Comprehensive antimicrobial treatments were applied to exposed framing, wall cavities, and subflooring, successfully passing all building inspections.
- Building envelope and drainage reinforced: Structural ceiling repairs in affected zones were completed, and main roof drainage systems were cleared and restored reducing the risk of future water back-up during severe weather events.
- Core host infrastructure re-established: Flood-damaged server hardware was replaced with clean systems, restoring dedicated virtualization capacity and stable storage for primary line-of-business workloads.
- Network authentication and domain services restored: Centralized network security and user access systems were re-established across redundant hardware, eliminating the single point of failure that had emerged during the disaster.
- Business-critical data fully recovered: Core accounting, bookkeeping, corporate tax records, payroll systems, and prospective client lists were completely recovered and verified for integrity. No operational data loss occurred as a result of the flooding event.
- System resilience and failover verified: Automated backup systems, server failover capabilities, and host storage layouts were tested. These tests confirmed that the restored infrastructure provides stable, predictable, and fault-tolerant operation suitable for long-term business continuity.

3.2 Lessons Learned and Recommendations

This project revealed several insights and improvement opportunities for future operational resilience.

- Roof drainage systems require proactive, scheduled maintenance: The flooding demonstrated that blocked gutters and downspouts can rapidly escalate water intrusion during severe weather. Regular inspection and cleaning of roof drainage systems are essential to prevent stormwater accumulation.
- Roof shingles and roof coverings must be kept in good repair: deteriorated shingles allowed rainwater to saturate ceiling insulation, causing structural damage and mold risks. Regular inspections and timely repairs are essential to protect physical premises.
- Primary domain controller health must be verified before starting any secondary domain controller: Bringing a secondary server online while the primary system is still experiencing errors can spread issues across the entire network. Recovery procedures must require fixing and testing the main server first to ensure a stable, error-free setup.
- Physical IT infrastructure elevation mitigates environmental exposure: Floor-level server hardware remains highly vulnerable to localized water inundation during physical facility emergencies. Corporate infrastructure guidelines should require elevating all core server chassis, network hardware, and power distribution units off the floor.
- Documentation must be maintained as a core operational asset: Clear, up-to-date guides for systems and recovery procedures proved essential during the restoration effort. Maintaining a reliable documentation library supports fast disaster recovery, new staff onboarding, and long-term infrastructure planning.
- Trade contractor sequencing requires dedicated daily oversight: Managing concurrent trades (drywall, plumbing, electrical, and flooring) during a regional emergency requires day-to-day coordination. Consistent oversight prevents scheduling conflicts, avoids rework, and keeps the overall project timeline on track.

4. Infrastructure Summary

This chapter presents a consolidated overview of the physical and technical infrastructure supporting corporate operations. It outlines the building layout, office environment, and supporting facilities, along with the IT systems that enable business-critical services and daily workflow across the organization.

4.1 Building Layout and Physical Environment

Corporate operations are distributed across a primary multi-level facility and an adjacent workshop used for building maintenance and repair activities. The layout is designed to support both administrative workflows and physical asset management, with work areas, storage rooms, and technical spaces arranged to support daily operations. Located in northeast Edmonton, the buildings are situated in one of the areas most severely affected by the 2026 extreme weather events, and several sections, particularly the basement office and the adjoining workshop, experienced significant flooding impacts.

4.1.1 Facility Overview

The company's physical footprint is split across two distinct structures on a multi-use property shared with other occupants. Within the main multi-level building, the company occupies three main-floor offices housing administrative workspaces and the primary server room, along with two basement rooms used for secondary workstations and a supply stockroom. Located approximately twenty feet away on the same property, a standalone workshop serves as a dedicated maintenance hub, housing workbenches and power tools used for facility upkeep and structural repairs across the site.

4.1.2 Main Level Layout

The main level serves as the primary operational hub, housing technical infrastructure and administrative workspaces across three distinct rooms. The dedicated server room contains core virtualized host servers, central network switches, and edge routing equipment. It also functions as the physical network entry point, where an external internet service provider coaxial line connects to a primary modem. To protect communications during power interruptions, a dedicated uninterruptible power supply supports the modem and routing hardware exclusively, while the server hosts manage power events independently. Physical distribution of key server hardware across different areas of the facility ensures domain fault tolerance and failover resilience.

The remaining space consists of two administrative offices supporting daily business activities. One functions as a standard private office, while the second is a larger shared workspace used by company personnel and other building occupants. Workstations in both offices connect through internal cabling directly to the server room to maintain secure access to central network services.

4.1.3 Basement Level Layout

The basement level contains six rooms, a central hallway, and a stairwell shared by multiple building occupants. The organization uses two dedicated rooms for secondary operations, long-term storage, and localized technical hosting. One room functions as an auxiliary office with workstation access to central network resources. The adjoining stockroom holds spare parts, maintenance materials, archived records, battery charging stations, and virtualized server hardware supporting lower-level workloads.

Because the basement sits partially below ground level, it is the most vulnerable area during severe weather. During the extreme rainfall of July 2026, water entered along the western perimeter and spread across both occupied rooms. A subsequent inspection identified several structural weaknesses that may have contributed to the intrusion, including aging window frames, exterior drainage limitations, and roof gutter discharge patterns. These combined factors made the basement office and stockroom the most affected areas of the primary building and required immediate remediation.

4.1.4 Adjoining Maintenance Workshop

The workshop is a standalone structure located roughly thirty to forty feet from the main building on the shared property. It serves as the primary maintenance area, containing heavy workbenches and a wide range of manual and power tools used for repairs and general upkeep. Separating these activities from the main building prevents noise and tool use from disrupting administrative workspaces and technical systems.

The workshop features an open floor plan used for large assembly, fabrication, and repair tasks. During severe rainfall in July 2026, water-soaked insulation caused a partial ceiling collapse that temporarily limited access and required structural remediation. The space is now fully restored and continues to support flexible maintenance needs for company personnel and other site occupants.

4.1.5 Intra-Office Connectivity

Intra-office connectivity is straightforward and minimal. The building's main and basement levels are connected by a single stairwell that provides access between floors. The main level has multiple entry points, and offices are linked by a standard hallway corridor. This simple, conventional layout supports routine movement and equipment transport. It also facilitated straightforward emergency response and cleanup during the 2026 extreme weather event.

4.1.6 Property Perimeter and Flood Impact Summary

The 2026 rainfall highlighted drainage issues around the building’s perimeter. To reduce future water intrusion, the inspector recommended improving the grading to direct water away from the building, extending the downspouts to carry runoff further from the foundation, and considering the replacement of the aging basement window if the budget allows. These measures will enhance the property’s exterior protection.

4.2 IT System Infrastructure and Network Topology

The organization relies on a centralized IT network to maintain reliable communication, store operational data, and support core business applications. These systems are distributed across physical servers and virtualized workloads that safeguard critical information and enable daily operations. Understanding the arrangement of server hardware, network routing, and central domain management provides essential context for how the technical environment functioned during the 2026 extreme weather events and how full system operations were restored during the recovery effort.

4.2.1 Physical Server Infrastructure

The organization's physical servers form the backbone of its IT environment. Primary server hardware is housed in the main-floor server room, consisting of several physical computers

operating as dedicated hypervisors to support all virtualized business servers. Currently, one hypervisor also functions as a print and scan server connected to a Canon Pixma printer and a standalone Canon scanner. Long-term plans include migrating these duties to a dedicated physical print and scan machine.

In addition to host hardware, the server room contains the internet modem along with core network routers and switches. To maintain critical internet connectivity during power outages, a dedicated uninterruptible power supply supports the modem and primary router.

Prior to the 2026 severe weather and flooding event, secondary server hardware operated in the basement workspace to distribute server loads across physical zones. Flood damage resulted in the total loss of the basement server hardware. Following post-flood remediation of the basement space, a replacement server will be deployed to restore the original dual-zone host configuration. Distributing physical host hardware across both levels preserves system redundancy and protects core operational assets.

4.2.2 Virtualized Workloads and Distribution

The organization uses a Hyper-V based virtual machine architecture to host all business servers and applications. Each physical hypervisor runs a set of virtual machines that provide core services such as authentication, file storage, printing, scanning, application hosting, and business operations. Workloads are allocated across the available hosts to maintain stability, reduce resource contention, and improve overall resilience.

Virtual machines are assigned to physical hosts based on performance needs, operational roles, and redundancy requirements. Critical workloads are distributed so that no single host carries all essential services. This approach ensures that if one physical computer becomes unavailable, remaining hosts can continue to support key business functions.

The virtualized environment also allows flexible scaling and simplified maintenance. New workloads can be deployed as additional virtual machines without requiring new physical hardware, and existing workloads can be moved or rebalanced across hosts as needed. This strategy supports long term growth while maintaining a stable and predictable operational footprint.

4.2.3 Network Routing and Connectivity

In this section, we detail the network routing and connectivity infrastructure for the facility. Internet connectivity enters through a dedicated coax cable, which is routed through the main

floor server room. This cable terminates at the central modem, which processes both static and dynamic IP addresses.

The static IP address is dedicated to the core business network, and the dynamic IP address supports non-critical wireless access. All core business systems utilize a wired router for enhanced security, ensuring that ethernet connections are used for primary business operations. All workstations on both the main floor and basement are connected via structured ethernet links routed through internal walls and conduits.

Additionally, a separate isolated router configuration is used for wireless devices, which are strictly separated into isolated subnets. This design helps ensure that non-essential wireless traffic remains separate from critical business operations and server communications.

4.2.4 Domain and Access Management

The company manages all network identities and authentication through a centralized Active Directory domain. To ensure continuity during unexpected outages, directory services operate across two domain controllers placed on separate floors of the building—one on the main level and one in the basement. This physical separation proved effective during the basement flooding incident, where the main-floor domain controller remained fully operational and continued to process user logins while the secondary system was restored.

Account permissions follow a deliberately simple two-tier structure aligned with the organization's size and trust model. A small administrative group retains full access to system settings, financial records, and sensitive operational data. All remaining staff use standard user accounts, which provide access only to the files, applications, and resources required for daily work.

To maintain long-term stability and reduce administrative complexity, the domain controllers use minimal customization and rely mainly on default Active Directory behaviors. The environment avoids extensive Group Policy Objects, reducing configuration overhead and simplifying troubleshooting. The primary exception is the Windows Server Update Services infrastructure, which applies local group policy and scheduled scripts on each workstation to handle registration, server-side targeting, and automated update delivery. This localized method keeps maintenance consistent across all systems without adding complexity to the domain controllers.

This streamlined domain architecture provides reliable authentication, clear permission boundaries, and a maintainable foundation for future growth while preserving operational simplicity.

4.2.5 Data Backup and Recovery Protocols

In our small company, we utilize a practical backup strategy designed around our limited resources and busy schedules. We operate two file servers, each on separate hypervisors to optimize disk space rather than for redundancy. These servers host active application data, line-of-business records, and daily user files across a combined capacity of approximately two terabytes.

Our primary backup process uses a custom PowerShell script that runs weekly to copy frequently changing operational data to a dedicated backup share. For static data that rarely changes, we maintain a separate archive share, which also holds exports of virtual machines to support quick host redeployments if a system fails. However, because point-in-time exports can break Active Directory, domain controllers are excluded from static backups; instead, identity resilience relies on live replication between our two physically separated domain hosts.

We do not currently store off-site backups, and all backup media remains within the building. While this poses a risk in the event of a total site loss, this approach reflects our current operational priorities and aligns with our available administrative bandwidth.

4.2.6 Post-Storm IT Restoration Summary

The extreme weather events of 2026 caused significant disruption to the company's physical workspace and core IT systems, requiring a coordinated restoration effort across facilities, hardware, and network services. While the flooding primarily affected the basement level, the impact extended into several technical systems that support daily operations.

The most critical recovery work involved restoring the basement-level domain controller and re-establishing stable Active Directory replication. With the main-floor controller remaining online throughout the incident, user authentication and access continuity were preserved. Once the damaged hardware was replaced and brought back into service, extensive troubleshooting was required to correct replication failures, rebuild trust relationships, and return the domain to a fully synchronized state. This work ensured that identity services remained reliable and resilient following the event.

Several virtual servers and application hosts also required attention due to moisture exposure, power instability, or forced shutdowns during the storm. Systems were brought back online in a controlled sequence, with configuration checks performed to confirm service integrity. File servers, application hosts, and supporting infrastructure were validated to ensure no data corruption or operational inconsistencies had occurred.

The restoration effort also included verifying the company's backup and archival processes. Weekly backup routines were reviewed to confirm that critical operational data remained intact, and archived virtual machine exports were checked to ensure they could still serve as recovery baselines if needed. Although all backups reside on-site, the existing procedures proved sufficient for restoring systems affected by the flooding.

Overall, the post-storm recovery demonstrated the effectiveness of the company's simplicity-focused IT architecture. Physical separation of key systems, minimal configuration complexity, and clear operational boundaries allowed core services to be restored quickly and reliably. The incident reinforced the value of maintaining straightforward, maintainable infrastructure that can be recovered efficiently even under constrained conditions.

5. IT Recovery: Server Hardware and Domain Services

When the extreme weather events pushed our systems to the brink, it became clear that restoring stability would require more than routine fixes. In this chapter, we'll explore the practical steps we took to rebuild our server hardware and restore our domain resilience.

5.1 Storm Impact and Loss of the Titan Server

In the wake of extreme weather events, the physical server Titan, located in the flooded basement level of the building, suffered a critical failure that reshaped the recovery strategy.

To understand the severity of this loss, it helps to understand Titan's role in the environment. Titan was a physical server functioning as a dedicated hypervisor. Rather than running applications directly on its own operating system, Titan's sole purpose was to host and manage virtual machines, which are isolated digital servers running on top of Titan's physical hardware. Among these hosted virtual machines were several secondary workloads and, most critically, DC-02, the network's secondary domain controller.

When storm runoff breached the basement utility area, the influx of water caused severe physical flooding and electrical instability. Titan was completely submerged and destroyed. When Titan failed, every virtual machine being hosted by Titan collapsed simultaneously. Beyond the immediate loss of application workloads, the destruction of DC-02 stripped the company of domain redundancy, leaving the entire organization dependent on a single surviving domain controller on the main floor. A single hardware failure on that remaining server would have brought all company network authentication to a complete halt.

While the physical host was destroyed, several virtual machine disk files remained safe. These workloads were hosted directly on a high-speed external solid-state drive connected to Titan. Unplugging this external drive preserved the intact virtual machine files, ensuring they would be available for immediate access once new host hardware could be commissioned. However, securing that replacement hardware proved far more complicated than expected.

Budget constraints dictated that the company could not purchase new operating system licenses. The recovery plan required reusing Titan's existing Windows Server 2019 Datacenter retail license. Staff initially attempted to deploy this license onto newly purchased retail desktop computers, but modern PC hardware presented an unexpected technical roadblock. Modern motherboards rely on updated storage controller drivers and BIOS configurations that are completely unrecognized by legacy Windows Server 2019 installation media. During setup, the installer simply could not see internal storage drives, blocking every attempt to install the operating system.

After multiple failed setup attempts on modern retail PCs, we shifted strategy. A refurbished, five-year-old server host was sourced, built on hardware from the same era as the Server 2019 media. This system fully recognized the storage controllers right out of the box, allowing Windows Server 2019 Datacenter to install cleanly. This refurbished host was commissioned as Mars, successfully establishing the physical hardware foundation for the next phase of recovery.

5.2 DC-01 Remediation and Primary Domain Health

With DC-02 destroyed in the basement flood, the organization was forced to operate its entire domain on a single surviving domain controller.

This created a high-stakes scenario for the systems administrator. With no backup domain controller to share the load or take over in the event of a crash, DC-01, located safely on the main floor, became the sole anchor for all user authentications, network permissions, and core directory services. Ensuring the continuous health and stability of DC-01 was paramount while the search for replacement server hardware was underway.

During this period we conducted thorough health checks on DC-01 to verify that active services were operating normally. Active Directory Domain Services, DNS, and user authentication continued to function smoothly for day-to-day business operations. The server successfully handled user sign-ins and granted access to network resources without interruption, reassuring management that primary operations remained secure on the single controller for the time being.

At this stage of the recovery, DC-01 appeared fully healthy and capable of carrying the network independently. However, because no secondary domain controller was present to attempt directory synchronization, the dormant replication faults appearing in the background remained hidden. Although DC-01 continued to operate normally, the domain was functioning without any safety margin; a single failure would have resulted in a complete authentication outage. The

systems administrator focused on maintaining DC-01 as a stable operational baseline while concurrently pursuing the replacement hardware required to re-establish domain redundancy.

5.3 Provisioning the Replacement Domain Controller

Following the severe weather event and the total loss of the Titan host machine, we purchased a refurbished Dell OptiPlex 7060, which we renamed Mars, to take over the crucial workloads that Titan used to run.

To ensure Mars had sufficient horsepower to replace Titan, the refurbished Dell OptiPlex 7060 was equipped with 64 gigabytes of DDR4 RAM and a 1.8 terabyte NVMe solid-state drive. This provided high-speed storage and enough memory headroom to host multiple virtual machines simultaneously without hitting resource bottlenecks. Once the hardware was unboxed and connected, Windows Server 2019 Datacenter installed cleanly onto the system. The installation was immediately activated using Titan's reused Datacenter license, satisfying OS compliance requirements without incurring additional licensing costs.

After the initial installation, the operating system's default random computer name was overridden and officially renamed Mars. With the base operating system stabilized and licensed, Mars was ready to join the network hypervisor infrastructure, functioning alongside existing hosts to manage dedicated virtual workloads.

Commissioning Mars with a Windows Server Datacenter license unlocked a key architectural benefit: Automatic Virtual Machine Activation (AVMA). Under this capability, any Windows Server virtual machines hosted on top of Mars inherit activation directly from the underlying Datacenter host without requiring individual product keys, up to and including Windows Server 2019. This established an efficient, scalable foundation for restoring Titan's former server workloads and provisioning new server instances as resources allowed, while client virtual machines, such as the Windows 11 TITAN-DOC workstation, continued using their respective desktop licensing.

The key discovery with AVMA is that AVMA client keys are generic public keys provided directly by Microsoft, but they must be injected via command line using slmgr while Hyper-V Data Exchange services are enabled.

AVMA Activation Steps

1. Host Verification: Ensure the physical Hyper-V host (Mars) is running an activated Windows Server Datacenter edition.

2. VM Setup: Boot the VM installer ISO, select "I don't have a product key" when prompted during setup, and complete the standard installation.

3. Integration Check: Verify that Data Exchange (Key-Value Pair / KVP exchange) is enabled under Hyper-V Integration Services (enabled by default) to allow hypervisor-level activation communication without network routing.

4. Command-Line Injection: Open PowerShell or Command Prompt as Administrator inside the guest VM and execute:

```
slmgr /ipk <AVMA-KEY-FOR-YOUR-EDITION>
```

e.g., using Microsoft's published Server 2019 AVMA keys such as TNK62-RXVTB-4P47B-2D623-4GF74 for Standard, or H3RNG-8C32Q-Q8FRX-6TDXV-WMBMW for data center.

Result: The VM immediately binds its activation directly to the underlying Hyper-V host and shows as permanently activated (slmgr /ato or checking System Properties).

Reference: https://learn.microsoft.com/en-us/windows-server/get-started/automatic-vm-activation?tabs=server2019

5.4 Provisioning and Attempting DC-03 (First Attempt)

With Mars fully provisioned as a hypervisor, the next step was to create a new virtual machine and promote it to a secondary domain controller to restore domain redundancy.

We created a fresh Windows Server 2019 virtual machine on Mars, configured its network settings, and initiated the standard domain join process. The virtual machine joined the existing Active Directory domain without incident. When we ran the Active Directory Domain Services configuration wizard to promote the server as DC-03, the process completed smoothly, and Server Manager confirmed that DC-03 was now an active domain controller.

However, operational tests quickly revealed that DC-03 was not functioning properly and failed to handle user authentication requests. Further inspection showed that the SYSVOL replication folders on DC-03 were completely empty, indicating that domain data had failed to replicate from DC-01. We ran manual commands to force directory synchronization:

```
repadmin /syncall

dfsrdiag SyncNow /Partner:"DC-01" /RGName:"Domain System Volume"
```

Despite executing these force-sync requests, the data would not traverse. To investigate the root cause, we ran the following diagnostic check on DC-03:

```
repadmin /showrepl
```

We executed this from the command line, expecting to see active directory replication links and successful inbound sync operations. Instead, the command returned no valid replication data. Following up on these results, we inspected the file system on DC-03 using both File Explorer and command-line navigation

```
dir C:\Windows\SYSVOL\sysvol
```

We used this check to verify the SYSVOL structure, only to find completely empty directory folders where synchronized policies and scripts should have resided.

Recognizing the risk of corrupting our only stable domain controller through unverified troubleshooting commands, we halted direct intervention on DC-01. Operating under the assumption that an error or missed step had occurred during the setup of DC03, we decided to abandon the virtual machine. We formally decommissioned DC-03, removed it from the domain, and prepared to attempt a clean installation with a new domain controller.

5.5 Provisioning and Attempting DC-04 (Second Attempt)

Following the decommissioning of DC-03, we built another virtual machine, named DC-04, to determine whether a clean installation would successfully establish domain controller replication.

DC-04 was created using the same Windows Server 2019 installation .ISO file in Hyper-V as its predecessor, with each step carefully verified to ensure accuracy. After activation via AVMA keys and a successful domain join, we promoted DC-04 to a domain controller. Once again, the promotion completed smoothly, and DC-04 appeared operational.

Yet the same replication issues persisted. DC-04 failed to receive any SYSVOL data from DC-01, and diagnostic checks revealed no active replication links. To investigate further, we ran a DFSR state query on DC-01 to determine whether it was accepting inbound replication. The command used was:

```
Get-WmiObject -Namespace "root\MicrosoftDFS" -Class "DfsrReplicatedFolderInfo" | Select-
Object ReplicatedFolderName, State
```

This check revealed that DC-01 was reporting a state value of 2 for the Domain System Volume, indicating that it was refusing replication attempts. This confirmed that the failure was not caused by procedural errors during the setup of DC-03 or DC-04; instead, the underlying issue originated from DC-01 itself.

Recognizing that continually deleting and recreating new domain controllers would lead to an endless cycle of failed attempts, we did not delete DC-04. Instead, we shut it down and

suspended all further testing on both DC-01 and DC-04. DC-01, hosted on Sirius, remained the only functioning domain controller, and any attempt to modify or repair it carried an unacceptable risk to the entire network. All live experimentation was halted to preserve domain stability.

5.6 Creation of the Isolated Sandbox Environment

Because any direct changes to our primary domain controller carried unacceptable risk, we set up a virtualized isolated sandbox in Hyper-V. This allowed us to perform controlled testing on cloned copies of the live domain controllers without touching the live environment.

After establishing the need for a safe testing environment, the first step was to create a new internal virtual switch named "internal-sandbox" on the Mars hypervisor. This switch provided a fully isolated network segment with no physical uplinks, ensuring that machines attached to it could communicate with one another while remaining completely detached from the live production network.

To establish a consistent and repeatable testing baseline, we performed an orderly shutdown of the relevant production virtual machines: the primary domain controller (DC-01), the secondary domain controller (DC-04), and a domain-joined workstation (WIN11PRO-1). Once powered down, we copied their VHDX virtual hard disk files into a dedicated baseline repository on Mars, which served as our file-level “gold master” reference state.

Whenever a test scenario concluded or required a fresh attempt, we simply powered down the sandbox VMs, deleted their modified VHDX files, copied fresh versions from the baseline repository, and reattached them through the Hyper-V Settings interface. This allowed us to reset the sandbox quickly while preserving all VM hardware configurations and switch bindings.

5.7 Resolving Domain Controller Interdependence and Sandbox Validation

In computing, a deadlock occurs when two parallel processes are mutually dependent on each other to proceed, leaving both permanently stuck in an unresolvable wait state; during initial testing, this exact deadlock condition occurred between our two domain controllers.

5.7.1 Deadlock Discovery and Root Cause Analysis

What first appeared to be a routine SYSVOL recovery quickly revealed a deeper problem: both domain controllers were influencing each other's failure state, creating a deadlock that neither could escape on its own.

DC-04 began in a neutral, uninitialized state. It was originally brought online in the production environment alongside DC-01, which had remained continuously online as our sole operational domain controller. Because DC-04 had not yet received any initial replication data from DC-01, it did not show any meaningful state when queried, sitting essentially in a waiting mode. Realizing that the underlying replication issues remained unresolved, DC-04 was shut down in production and a baseline snapshot image was captured to construct its sandbox clone. Consequently, when deployed into the sandbox, DC-04 was still waiting on DC-01 to push initial replication data before it could become operational.

To restore DC-01 to an authoritative state, execute the following steps on DC-01.

Remediation Steps

1. **Disable DFSR in Active Directory**
    - Open ADSI Edit (adsiedit.msc)
    - Set msDFSR-Enabled = FALSE
2. **Force AD Polling**
    - Run: dfsrdiag PollAD
3. **Set Authoritative Restore Option**
    - Open ADSI Edit (adsiedit.msc)
    - Set msDFSR-Options = 1
4. **Re-enable DFSR in Active Directory**
    - Open ADSI Edit (adsiedit.msc)
    - Set msDFSR-Enabled = TRUE
5. **Force AD Polling**
    - Run: dfsrdiag PollAD
6. **Restart DFSR Service**
    - Run: net stop dfsr

    - Run: net start dfsr
7. **Monitor State Transition**
    - Wait for DC-01 to enter State=4
8. **Verify SYSVOL Share**
    - Confirm \\dc-01\sysvol is shared

Under simultaneous boot conditions before performing the remediation steps, an immediate conflict developed because neither domain controller possessed a validated, authoritative copy of SYSVOL to offer the other. DC-04 was introduced into the environment in an uninitialized state, relying entirely on DC-01 to push initial replication data. However, DC-01 was itself sitting in State=2 (Initial Read / Un-initialized) and could not serve as a replication source until its authoritative restore procedure was completed.

This setup immediately triggered a mutual dependency loop between the two servers. Upon booting, DC-04 polled DC-01 to establish its baseline SYSVOL state. Detecting that DC-01 was reporting State=2, DC-04 automatically set its own local status to State=2 to prevent bad or unverified replication data from corrupting its database.

After the remediation steps were executed on DC-01, DC-01 polled its active replication partner, DC-04, as part of its initialization check. Detecting that DC-04 was now reporting State=2, DC-01's DFSR service interpreted partner participation as an active conflict. Rather than completing its transition to State=4 (Normal), DC-01 aborted its recovery routine and reverted back to State=2.

This created a circular wait deadlock when both servers were running together:

- **DC-04 polling DC-01** → DC-04 entering State=2
- **DC-01 polling DC-04** → DC-01 aborting recovery and reverting to State=2

Simultaneous booting directly triggered this deadlock because both DFSR services were active and polling each other concurrently. DC-04 continuously signaled an error state back to DC-01, ensuring that any attempt to execute the remediation steps on DC-01 would be aborted before authoritative initialization could complete.

The remediation steps were designed to let DC-01 complete its authoritative SYSVOL initialization before DC-04 attempts to synchronize. This sequence fails under simultaneous boot conditions because both domain controllers are actively polling each other from the very start. As a result, DC-01 cannot progress beyond State=2, since it constantly receives a conflict signal

from DC-04, which is equally in an uninitialized state. Thus, neither controller can complete its initialization phase because they remain interlocked in a cycle of dependency.

5.7.2 Corrective Strategy: Breaking the Deadlock

By simply reordering the configuration steps, we found a straightforward path to break the deadlock and restore stability.

- Isolation strategy: Keeping DC-04 offline
- Authoritative restoration of DC-01 to reach State=4 in isolation
- Delayed startup for DC-04 and automatic SYSVOL synchronization without manual ADSI Edits.
- Why this sequencing works
- How it prevents recurrence

This is the “how we solved it” subsection.

By simply reordering the configuration steps, we found a straightforward path to break the deadlock and restore stability.

The first requirement was to keep the secondary domain controller offline. This allowed the primary controller to complete its remediation sequence without receiving conflicting DFSR state information during initialization.

With the secondary controller powered off, the primary controller advanced cleanly through its authoritative restore and reached State=4. Once in this state, it held a fully validated SYSVOL dataset and was ready to serve as the replication source.

Only after the primary controller stabilized was the secondary controller brought online. At startup, it automatically performed a non-authoritative synchronization and replicated SYSVOL from the primary controller without re-doing the ADSI edits used in the primary controller’s remediation.

This sequencing works because it ensures the authoritative controller finishes its recovery uninterrupted. Once that controller is healthy, the secondary controller can safely initialize and synchronize, preventing the circular dependency from forming again.

5.7.3 Validation Methodology and Results Across Six Controlled Scenarios

Validating domain authentication is notoriously tricky on modern client operating systems, where hidden credential caching can easily disguise an ongoing failure as a success.

When a domain workstation successfully authenticates a user, Windows caches those credentials locally. If domain controller communication subsequently fails, the workstation falls back on cached credentials, granting interactive logon and making a broken domain connection appear operational. To establish true operational recovery, our testing required brand-new domain accounts with zero logon history on the test client, a sandbox clone of a Windows 11 workstation named WIN10PRO-1.

To isolate baseline failures from post-remediation successes, six unique test accounts were provisioned in Active Directory and granted Remote Desktop privileges. Each test consisted of a single interactive RDP logon attempt to WIN10PRO-1 to verify real-time Active Directory authentication.

To isolate baseline failures from post-remediation successes, six unique test accounts were provisioned in Active Directory. To ensure logon attempts failed only due to domain health rather than local authorization policies - avoiding false negative results - each test account was explicitly granted remote access permissions on WIN10PRO-1.

To configure remote desktop permissions on WIN10PRO-1:

1. Open Computer Management (compmgmt.msc).
2. Navigate to System Tools > Local Users and Groups > Groups.
3. Double-click the Remote Desktop Users group.
4. Click Add, click Locations, and select the domain directory.
5. Enter the target test account name (e.g., pre-14), then click Check Names to validate the account.
6. Click OK, then click Apply to save the changes.

This process was completed for each test account prior to execution. Each test then consisted of a single interactive RDP logon attempt to WIN10PRO-1 to verify real-time Active Directory authentication.

Prior to executing the remediation steps, baseline tests were conducted across three domain controller uptime states:

Scenario	**User ID**	**Result**
Both DCs Online	pre-14	Pass

DC-01 Only	pre-1	Pass
DC-04 Only	pre-4	Fail

The failure of pre-4 confirmed that DC-04 was unable to process domain authentication independently due to its uninitialized SYSVOL state.

Following the execution of the remediation steps—isolating DC-04, authoritatively restoring DC-01, and completing delayed boot synchronization on DC-04—the exact same test matrix was repeated with new post-fix accounts:

Scenario	**User ID**	**Result**
Both DCs Online	post-14	Pass
DC-01 Only	post-1	Pass
DC-04 Only	post-4	Pass

Transitioning the DC-04-only scenario from Fail (pre-4) to Pass (post-4) provided conclusive proof that the remediation was successful. SYSVOL replication was restored, DFSR state was healthy, and DC-04 was fully capable of independent domain authentication.

5.8 Restoring Domain Redundancy

After days of controlled testing and careful remediation, the moment of truth arrived. With the primary controller stabilized, the secondary came online, and the domain snapped back into full replication without a single error.

DC-04 completed its promotion sequence cleanly, and the expected SYSVOL and NETLOGON shares appeared without delay. Initial replication checks confirmed that directory data was synchronizing normally between DC-01 and DC-04, and repadmin output showed no lingering objects, no backlog, and no topology inconsistencies. The DFS Replication and Directory Service event logs reported standard initialization activity with no warnings or error conditions, indicating that the earlier issues encountered during the DC-03 attempt had been fully resolved.

Alongside core directory services, live data .vhdx virtual disk files from Titan's former virtual machines were successfully recovered and mounted on Mars. This included spinning up TITAN-DOC, a Windows 11 workstation environment equipped with Visio Pro 16, restoring immediate operational capability and setting the stage to draft system architecture diagrams for subsequent reporting. With additional legacy Titan VMs now staged for hosting on Mars, this restored the resilient, fault-tolerant baseline that had been missing since Titan's failure, bringing

a well-earned sense of relief as the core directory recovery effort was officially and successfully completed.

5.9 Restoration of Remaining Titan Virtual Machines

- Opening paragraph: More info about other things that were on Titan. Description of Titan as a critical multi-role hypervisor running eight essential virtual machines across domain infrastructure, line-of-business applications, developer environments, and databases.
- Recovered / Operational Workloads (2 VMs):
    - DC-02 / DC-04: Secondary Domain Controller (re-provisioned and fully synchronized).
    - TITAN-DOC: Windows 11 documentation workstation (recovered .vhdx, hosted on Mars, equipped with Visio Pro 16 for technical reporting).
- Pending / Staged Workloads for Recovery (6 VMs):
    - W10P-3: General-purpose Windows 10 Pro workstation environment.
    - W16-MNY98: Specialized application server (Money 98 / financial application host).
    - W16-QBB21: Application server dedicated to QuickBooks Desktop 2021 financial data.
    - Win2016-5: General-purpose Windows Server 2016 application host.
    - Titan-Dev-3: Primary .NET / Azure development workstation environment.
    - W16-CosmosDB: Database server hosting local Cosmos DB emulator/data instances.
- Priority Restoration:
    - Given the critical nature of certain systems, we will begin by restoring the three remaining servers W16-CosmosDB, W16-MNY98 and W16-QB21.
    - Then we will restore the developer workstation titan-dev-3 because it provides support for live cloud-based production applications.
    - And finally we will restore the remaining general purpose VMs.
- Strategic Hosting Plan:
    - Staging order and priority matrix for bringing these remaining six workloads online as memory and storage allocations permit across all available physical hypervisors (Lunar, Europa, Sirius, Mars). Mars brings a lot of processing power to the table making it the most powerful physical computer in the fleet. Currently the Sirius hypervisor is extremely overloaded, so much so that it cannot run all of its current workload of nine VMs at the same time. So some redistribution of all the VMs is a good idea at this point.

5.10 Resilience on a Shoestring: Pragmatic Infrastructure Hardening

With facility funds prioritized for urgent roof and window frame repairs as well as structural remediation, achieving IT physical infrastructure resilience became an exercise in practical ingenuity.

Rather than waiting for enterprise server racks or dedicated enclosures, protective measures were implemented immediately using existing on-site materials. To eliminate the ground-level vulnerability that claimed Titan, the Mars hypervisor host is elevated using its own heavy-duty shipping box. Sealed securely and repurposed as a temporary staging base, this improvised riser provides reliable vertical clearance until permanent shelving or furniture is acquired. This low-cost physical lift approach guarantees that future water intrusion from unexpected window or foundation leaks will pass harmlessly beneath the hardware without disrupting active hypervisor workloads.

Taken together, these adjustments demonstrate that meaningful IT resilience does not depend on large budgets or specialized equipment. Through practical hardware placement, disciplined recovery work, and targeted environmental safeguards, the environment now operates with greater stability and redundancy than it had prior to Titan's failure. The recovery effort has restored core services, strengthened physical safeguards, and established a reliable foundation for future improvements as resources permit. Effective IT hardening is being achieved through thoughtful, resource-efficient decisions that strengthen critical systems and support long-term operational reliability.